Catherine King

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PROFILE

Information and computational social scientist researching interventions that address online harms, focusing on both effectiveness and user acceptance. Recent work identifies failure modes in AI-generated search summaries and proposes technical and legal policy solutions. Methods include network science, social media analytics, surveys, and user studies.

Postdoctoral Fellow in Policy and Special Faculty in the School of Computer Science, Carnegie Mellon University.

Research interests: online harms and misinformation; technology and AI policy; social computing; AI-mediated information; user acceptance; public opinion

EDUCATION

Carnegie Mellon University – *Ph.D. in Societal Computing* *2025*
Software and Societal Systems Department
Dissertation: *Effective and Practical Strategies for Combatting Misinformation*
Committee: Kathleen M. Carley (Chair), Hong Shen, Chris Labash, Pablo Barberá

The College of William & Mary – *M.S. in Computational Operations Research* *2016*
Capstone: *Optimizing voter wait times* (simulation and paper)
Advisor: Lawrence M. Leemis

The College of William & Mary – *B.S. in Mathematics, Minor in Computer Science* *2014*
Summa cum laude, Honors in Mathematics
Thesis: *Nonlinear models of zooplankton communities*
Committee: Sarah Day (Advisor), Drew LaMar (Advisor), Mark Brush, Lawrence M. Leemis

SELECTED PUBLICATIONS

Preprints

- **King, C.,** Ng, L. H. X., and Carley, K. M. (2026). Beyond Effectiveness: A Multi-Criteria Framework for Comparing Practical Socio-Technical Interventions. *Submitted to CHI 2027.* arXiv:2608.20649 [cs.AI]

Peer-Reviewed

- **King, C.,** Ng, L. H. X., and Carley, K. M. (2026). User-Reported Misinformation Exposure Across Social Media Platforms. *Proceedings of the 2026 SBP-BRiMS Conference.* [Paper]
- **King, C.,** Phillips, S. C., and Carley, K. M. (2026). Public Support for Misinformation Interventions Depends on Perceived Fairness, Effectiveness, and Intrusiveness. *Journal of Online Trust and Safety, 3*(2). [Paper] [Blog]
- **King, C.** and Carley, K. M. (2025). Promoting Social Corrections: A Media Literacy Intervention for Misinformation on Social Media. *Proceedings of the 2025 SBP-BRiMS Conference.* [Paper][Blog]
- **King, C.,** Carragher, P., and Carley, K. M. (2025). Mapping the Scientific Literature on Misinformation Interventions: A Bibliometric Review. *COMPASS 2025: ICWSM Workshop Proceedings.* [Paper]
- **King, C.,** Phillips, S. C., and Carley, K. M. (2025). A path forward on online misinformation mitigation based on current user behavior. *Scientific Reports, 15*. [Paper][Blog][News Article]
- Moffitt, J. D., **King, C.,** and Carley, K. M. (2024). Connecting the Domains: An Investigation of Internet Domains found in COVID-19 Conspiracy Tweets. *Computational and Mathematical Organization Theory, 30*. [Paper]

- **King, C.** and Carley, K. M. (2023). Gender dynamics on Twitter during the 2020 U.S. Democratic Presidential Primary. *Social Network Analysis and Mining, 13*(50). [Paper][Blog]
- Moffitt, J. D., **King, C.,** and Carley, K. M. (2021). Hunting Conspiracy Theories during the COVID-19 Pandemic. *Social Media + Society, 7*(3). [Paper][Blog]
- Bellutta, D., **King, C.,** and Carley, K. M. (2021). Deceptive accusations and concealed identities as misinformation campaign strategies. *Computational and Mathematical Organization Theory, 27*. [Paper]
- **King, C.,** Bellutta, D., and Carley, K. M. (2020). Lying About Lying on Social Media: A Case Study of the 2019 Canadian Elections. *Proceedings of the 2020 SBP-BRiMS Conf.* **[BEST PAPER AWARD]** [Paper]
- Glasser, S. and **King, C.** (2019). System dynamics for estimating sUAS operations. *Proceedings of the 2019 Winter Simulation Conference.* [Paper]
- **King, C.** and Leemis, L. M. (2016). Data analysis and simulation: Optimizing voter wait times. *Proceedings of the 2016 IEEE Systems and Information Engineering Design Symposium.* [Paper]

Technical Reports

- **King, C.,** Lepird, C. S., and Carley, K. M. (2021). Project OMEN: Designing a Training Game to Fight Misinformation on Social Media. Carnegie Mellon University, Institute for Software Research.

FELLOWSHIPS AND AWARDS

- **Outstanding Service Award (2026):** For leadership in organizing the 2026 IDeaS Conference
- **Knight Fellow (2020 – 2025):** Transdisciplinary research fellowship (project yielded two papers: *Scientific Reports* and *Journal of Online Trust and Safety).*
- **Best Paper Award (2020):** SBP-BRiMS Conference, *Lying about lying on social media*
- **"Fan Favorite" Award (2016):** MODSIM World conference competition; runner-up for the judges' award (https://faster-voting.wm.edu/).
- **3rd place, WINFORMS Student Excellence Award (2016):** original research in Operations Research.
- **Luther T. Connor Prize (2014):** Mathematics department award, determined by faculty.
- **1st place, Undergraduate Research Competition (2013):** International Symposium on Biomathematics and Ecology Education and Research conference (presentation and manuscript).

RESEARCH EXPERIENCE

Carnegie Mellon – Postdoctoral Fellow in Policy and Special Faculty *2025 – present*

- Lead a research project on Google AI Overviews, cataloging failure modes across each layer of its retrieval-augmented generation (RAG) pipeline, analyzing the U.S. legal landscape, and proposing technical and legal policy solutions
- Lead the policy program at the Center for Informed Democracy and Social-cybersecurity (IDeaS), which includes organizing the 2026 IDeaS Conference and inviting seminar speakers
- Supervise a Master's student intern on the AI Overviews project

Carnegie Mellon – PhD Researcher *2019 – 2025*

- Designed and fielded surveys on social media user behavior and perceptions of misinformation countermeasures
- Co-developed Project OMEN, a training game for government analysts to improve misinformation detection and countering abilities
- Created and tested a media literacy intervention, evaluated effectiveness via a controlled user study
- Conducted a bibliometric analysis of 450+ misinformation intervention papers
- Quantified gendered dynamics and hate speech toward U.S. presidential candidates in 2020

William & Mary – Research Assistant *2012 – 2014*

- Modeled zooplankton-phytoplankton dynamics in the Chesapeake Bay
- Presented research at multiple mathematics conferences
- Resulted in an Honors thesis in Mathematics

TEACHING EXPERIENCE

Carnegie Mellon – Teaching Assistant *2019 – 2025*

- *Dynamic Network Analysis* (graduate level) – Led recitation sessions for 1 semester.
- *CASOS Summer Institute* – Delivered modules (2020-2024) on hashtag community analysis, the OMEN exercise, one- and two-mode metrics and folding, and network comparison/prediction.
- *OMEN Exercise* (2024) – Led training sessions on misinformation detection and countering.

2013 – 2016

William & Mary – Teaching Assistant / Grader

- TA for *Calculus II* (4 semesters) and *Introduction to Mathematical Biology* – led labs and office hours
- Grader for multiple advanced mathematics courses, including *ODEs* and *Mathematical Statistics.*

PROFESSIONAL EXPERIENCE

MITRE – Data Scientist *2018 – 2020*

- Built spectrum consumption models and conducted analysis in Java.
- Co-developed a system dynamics model forecasting UAV adoption rates in the U.S.
- Supported Army manpower modeling efforts via linear regression models and economic analyses.

Genworth – IT Analyst / Modeling Analyst *2016 – 2018*

- Maintained and developed actuarial models in MATLAB, SAS, and R
- Implemented long-term care projection simulations in MATLAB

ACADEMIC AND COMMUNITY SERVICE

- **Peer Reviewer** – *Nature Human Behaviour, Scientific Reports, New Media & Society, JMIR Infodemiology, Social Media + Society, Policy Studies Journal, Behavioral Sciences of Terrorism and Political Aggression,* SBP-BRiMS, and IDeaS Conference
- **Guest Co-Editor (forthcoming):** "Governing the Digital Age: AI, Platforms, and Online Harms Policy," special issue of *Computational and Mathematical Organization Theory*
- **Organizing Committee Lead, 2026 IDeaS Conference:** Oversaw submission review, selected and invited panelists, and selected the best paper winners. Won the **[OUTSTANDING SERVICE AWARD].**
- **Mentor (2024)**
  - Managed three high school interns in a directed reading program
  - Trained interns to categorize misinformation intervention types
- **CMU Graduate Student Association (GSA) Representative (2023 – 2025)**
  - Planned and budgeted for social events for students in the Societal Computing Ph.D. program
  - Liaised with department leadership and advocated for student issues
- **CMU GSA, External Affairs Committee Member (2020 – 2025)**
  - Recruited >20 CMU students to be poll workers during the 2020 poll worker shortage
  - Met with congressional staff on graduate student issues (e.g. "CHIPS and Science Act")
  - Registered students to vote in advance of the 2022 elections
- **Election Judge / Poll Worker, Pittsburgh (2020 – present)**
  - Manage election day operations and supervise poll workers
  - Ensure ballot security and fairness, and certify the precinct's results

SKILLS

Technical: R, Python, ORA, MATLAB, LaTeX (proficient); Java, SQL, SAS (familiar)
Methods: Social network analysis, bibliometric analysis, survey analysis, regression modeling, simulation
Languages: English (native), Italian (heritage)